

Sultan Of Oman
Ministry of Health
Royal Hospital



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Ministry of Health
The Royal Hospital
Department of Blood Transfusion Services

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Title: Transfusion in Neonates and children under six months of age

Compatibility testing in neonates:

- Abbreviated testing is adopted in neonates (less than one month of age) and infants (< 4 months of age) due to the fact that they have immature immune system and unlikely to have developed natural occurring antibodies (anti-A, anti- B. etc.) or irregular antibodies at this age.
- Antibodies which are detected at this age are probably of maternal origin, hence it is important to have access to maternal sample for testing. Therefore, for both neonates and infants less than 4 months of age the following tests are only done:
 1. Abbreviated ABO blood grouping (i.e. Forward grouping without reverse grouping)
 2. Direct anti-globulin test (DAT). If DAT is positive then further testing must be performed on both mother's and baby's blood sample including antibody screen.
 3. If the mother is known to have irregular antibodies against red cell antigens (alloimmunized), maternal sample should be provided for testing so an additional cross match test against the maternal sample must be performed to avoid haemolytic transfusion reaction in the neonate or infant (< 4 months of age).
- Mothers ABO / Rh group and antibody status must be known
- Blood samples of infants older than 4 months will be tested similar to adult's blood samples.

Red cell transfusions specifications:

- These should be given as packed red blood cells, group O, Rh-compatible
- Packed red cell units are less than 10 days old
- Sick cell trait is negative.
- If the same unit is requested to be used multiple times for the same patient, then the first time a unit is selected it must be 10 days old or less. For subsequent transfusions, the same unit can be used even if it is more than 10 days old.

- All aliquots will be placed in a Paed Pack (which can hold up to 75 ml of blood)
- Blood Bank Laboratory will add 10 ml to the requested volume (e.g. a request of 50 ml means transferring 60 ml of blood)

Reconstituted Whole Blood for Exchange Transfusions:

- These units are prepared in the blood bank by mixing one unit of **group O, Rh-compatible packed red cells** mixed with one unit of **group AB Rh-positive fresh frozen plasma** in a closed system.
- This constituted unit shelf life is 24 hours.
- If the mother is iso-immunized (i.e. **developed irregular antibodies against the child's red cells**). For example, anti-D, anti-c, anti-K, anti-Jka), then **antigen negative group O packed cells** unit are selected for preparation of the reconstituted whole blood unit.

Platelet Transfusions:

ABO-identical and Rh-compatible units **MUST** be selected.

Fresh Frozen Plasma Transfusions:

- The Blood Bank staff will thaw **group AB plasma**.
- Thawed units will be discarded after 4 hours, if not used.

Appendix: Choice of ABO group for blood components for children under six months of age

Patient's ABO Group			
	Red cells	Platelets	FFP *
O			
First Choice	O	O	O
Second Choice	-	A or B	A or B or AB
A			
First Choice	A	A	A
Second Choice	O	B	AB
Third choice	-	O	
B			
First Choice	B	B	B
Second Choice	O	A	AB
Third choice	-	O	
AB			
First Choice	AB	AB	AB
Second Choice	A or B	A or B	
Third choice	O	O	

Ideally **components** which test negative for “high titre” anti-A and / or anti-B should be selected but this test is not available. The use of group O platelets for non-group O patients should be avoided as much as possible.

Platelet concentrates of group B or AB may not be available.

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Review Details

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